7.8.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 7, Lesson 9: Using Trigonometry in the Real World

Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Target

- [ ] I can solve real-world problems using trigonometric ratios and the Pythagorean Theorem.

7.9.1 Warm-Up: Notice and Wonder

1. The picture shows the stairs in the Ponce de Leon Inlet Lighthouse.

   a. What do you notice?

   b. What do you wonder?

   c. The lighthouse has 203 steps. Do you think this picture includes all of the steps? Explain your answer.

7.9.2 Collaboration: Angle of Depression and Angle of Elevation

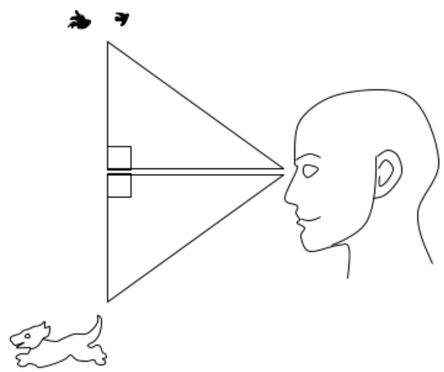
Real-world trigonometric ratio problems involve angles that are made from a horizontal line of sight.

1. Using a ruler, draw the horizontal line of sight for the man.

2. The horizontal line of sight is always ○ perpendicular ○ parallel to the ground.

3. The angles created from the horizontal line of sight are called angles of elevation and angles of depression. In the image, Hector is looking at birds in the sky and a dog on the ground.

   Label the angle of elevation with an E, label the angle of depression with a D.



4. Based on the name of the angle, work with a partner to write a definition for an angle of elevation.

5. Work with a partner to write a definition for an angle of depression.

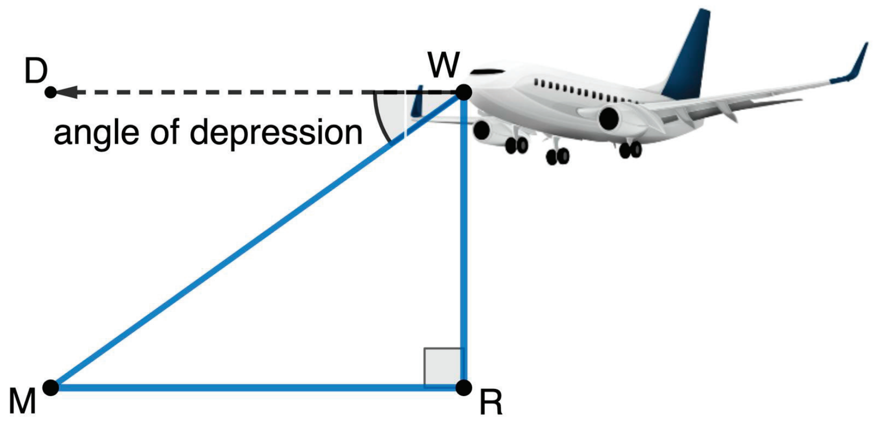
6. Compare your definitions with that of another pair. Discuss any differences. Make changes as necessary.

7.9.3 Guided Instruction: Angles of Elevation and Depression

The **angle of elevation** is the angle created above the horizontal when looking up at an object.

The **angle of depression** is the angle created below the horizontal when looking down at an object.

1. The angle of depression, $m\angle DWM$, from the airplane to the runway, point M, is $37°$.

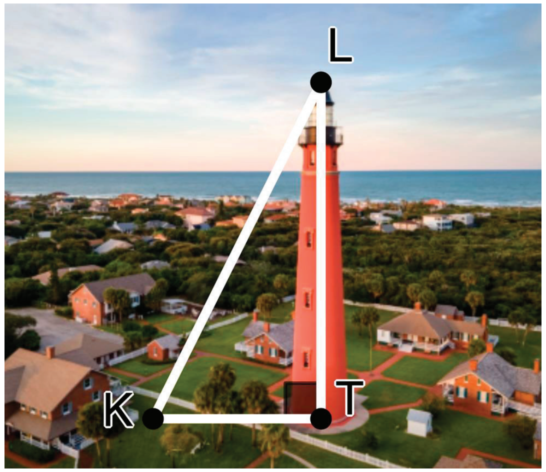


   a. Explain why $\angle DWM$ is congruent to $\angle RMW$.

b. If the airplane is flying at an altitude of 2500 feet, WR, and is descending to the airport, M, what is the distance, MW, the plane has to travel?

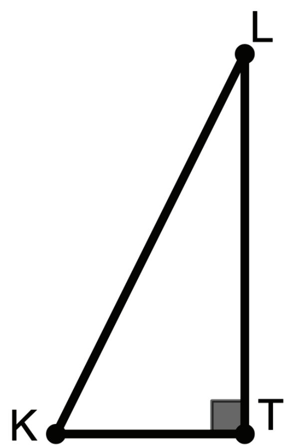
c. How far is the runway from a point on the ground directly below the airplane, MR?

2. Kamala is visiting the Ponce de Leon Inlet Lighthouse. Kamala is standing at point K and is looking at the top of the lighthouse, point L. $\overline{KT}$ is from Kamala's eye level, which is at 1.75 meters (m).



a. When Kamala looks at the top of the lighthouse, is the angle created an angle of elevation or depression?

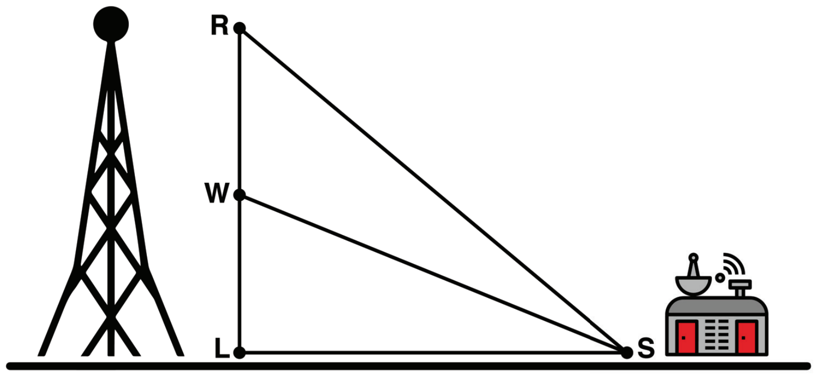
b. The height of the Ponce de Leon Inlet Lighthouse is 53 m. Kamala is standing 15.25 m from the lighthouse. Label ΔKTL appropriately with this information.



c. What is the measure of the angle created when Kamala looks at the top of the lighthouse?

7.9.4 Guided Practice: Trigonometry in the Real World

1. A cell phone tower was built in two sections, with a control shed built 60 feet from the base of the tower on level ground. From the base of the control shed, the angle of elevation, $\angle LSW$, to the top of the first section of tower, $\overline{LW}$, is $22°$. The angle of elevation to the top of the second section, $\angle LSR$, is $46°$.



a. Label the diagram with the information from the problem.

b. Determine the length of $\overline{RW}$, the second section of tower. Round your answer to the nearest thousandth of a foot.

2. A hot air balloon is flying 38.5 ft above level ground. The Orlando City Soccer field is 200 ft from the point on the ground directly below the balloon.

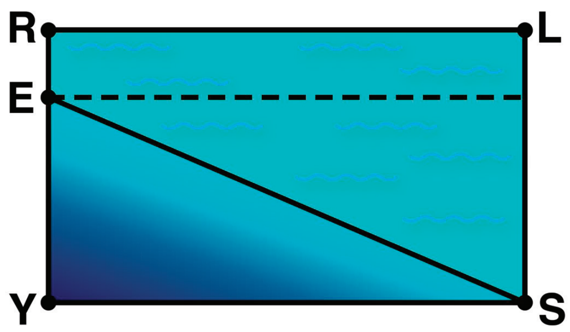
   a. Draw and label a diagram using the information from the problem.

   b. What is the angle of elevation from the soccer field to the hot air balloon?

   c. The hot air balloon ascends further into the air. If the new angle of elevation from the soccer field to the balloon is $17.3°$. How many feet did the hot air balloon rise?

7.9.5 Your Turn!

1. A swimming pool is 48 feet long and 4 feet deep in the shallow end. There is a steady decline of $14°$ toward the deep end of the pool. Rectangle $RLSY$ and ΔSYE are shown to model the transition from the shallow end to the deep end. Label the diagram appropriately.

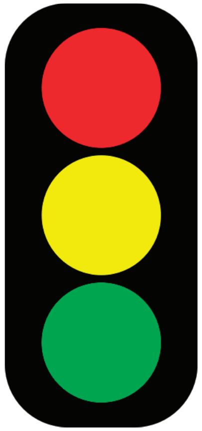


   a. How deep is the pool at the deep end?

   b. $\overline{ES}$ represents the slant in the pool from the shallow end to the deep end. What is its length?

7.9.6 Wrap-Up: Reflection

1. Circle the color on the stoplight that best represents your understanding of solving real-world problems using an angle of depression or an angle of elevation.



Red: I do not understand.

Yellow: I need some more help.

Green: I got this; I understand!

2. Complete the following statements if you selected red or yellow.

   a. I am still unsure about …

   b. To fill in this gap I intend to …

Unit 7, Lesson H1: Law of Sines

Benchmark

MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.

Learning Target

- [ ] I can solve mathematical and real-world problems involving triangles using the Law of Sines.